(vi) If sampling without replacement scheme is followed, a random number once recorded appears again, the same is omitted and we move on to the next number.

(vii) The 10 such selected random numbers are compared with the labeled (numbered) population units and the corresponding units are selected to get a simple random sample of size 10.

Example 2.1

Select a random sample of size 15 from the random number table from a finite population of size 220.

Step 1 : Assign serial numbers 00 to 219 to the 220 units of the population.

Step 2 : Since the maximum digit is 3, select a three digit as starting point. Let the starting point in the random number table is 066.(first two digits of the entry in the 11th row and 4th column).

Step 3 : Continue in the column downwards to select random numbers of size 15.

Step 4 : The numbers chosen to have SRS are 066, 147, 119, 194, 093, 180, 092, 127, 211, 087, 002, 214, 176, 063 and 176. If we get any random number greater than 219, then the multiple of 219 is to be subtracted from the selected number such that the resultant value is less than 219.

Illustration:

Assume that the random is 892. But here, the maximum value assigned for the population is 219 . Hence we select '4'as the multiplier of 219. Then the random number correspond to 892 is 016. (See: $892 - 4 \times 219 = 892 - 876 = 16$)

Case(1) : Under SRSWR: Here the random sample is 066, 147, 119, 194, 093, 180, 092, 127, 211, 087, 002, 214, 176, 063 and 176.

Case(2): Under SRSWOR: Here the random sample is 066, 147, 119, 194, 093, 180, 092, 127, 211, 087, 002, 214, 176, 063and 157.

[**Note:** 176 is removed from the selection since it is repeated again]

Merits:

- It is more representative of the universe.
- It is free from personal bias and prejudices.

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